

DEMAND AND SUPPLY EFFECTS OF UNIVERSITY
FUNDING CHANGES:

AN AUSTRALIAN POLICY ANALYSIS

MAXWELL YONG

ECON40018 & ECON40019
Economics Research Essay

Submitted in partial fulfillment of the requirements of
the Bachelor of Commerce (Honours).

October 2022

Department of Economics
University of Melbourne

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Signed: Maxwell Yong

Date: 17th of October, 2022

ACKNOWLEDGEMENTS

I would like to acknowledge the valuable support I received from my research advisers: Associate Professor Michael Coelli and Dr. Jan Kabátek. I have felt incredibly supported by their expert knowledge, unparalleled intuition, and endless reassurances. Moreover, I am grateful to The Life Course Centre (LCC) for supporting my research through the Life Course Centre Future Pathways Scholarship.

A big thank you is owed to Helen Tam and Anthony Manny from the Universities Admission Centre (UAC) who provided the incredibly rich administrative dataset I use for the basis of my analysis. Moreover, I am appreciative to the Melbourne Institute for lending me a Stata-equipped computer to allow me to run models non-stop for months.

I acknowledge Professors Andrew Norton and Bruce Chapman from the Australian National University (ANU) for their extensive help. I am grateful to Dr. Aaron Barkley who was happy to discuss supply-side estimation ideas for weeks on end *and* read a draft. Thank you also to A/Prof Victoria Baranov for discussing identification strategies.

Thank you to the attendees of the ANU HECS-HELP policy forum who provided me with useful insights and valuable feedback: A/Prof Janine Dixon, A/Prof Tim Higgins, Dr. Clare Hourigan (and her Planning & Business Intelligence team), Dr. Gareth Bryant, Dr. Tracey West, Mark Warburton, and academics from the Melbourne Centre for the Study of Higher Education (Dr. Gwilym Croucher, Professor Sophie Arkoudis, and Professor Chi Baik). I appreciate everyone who read a draft and gave me other helpful advice: Daniel Tiong, Dr. Jim Minifie, Dr. Kaely Woods, Dr. Daniel Avdic (from the Deakin University research competition), Manning Clifford, and my mum. Thank you to all the academics who discussed my various research ideas before the start of my Honours year: Professor Peter Dixon, Professor Roger Wilkins, Professor Mark Wooden, Professor Richard Holden, and Dr. Lawrence Uren.

A big thanks goes to everyone in the UniMelb Economics Honours program (overseen by the illustrious A/Prof Mei Dong). A special thanks to Dominic Merrified, Joel Fernando, Oscar To, and Felix Capp who kept me sane throughout the year and provided me with feedback on my research. Lastly, thank you to my family, friends, and my girlfriend for supporting me throughout the year and putting up with endless HECS-HELP discussions.

DISCLAIMER

This essay uses unit record data from the Universities Admissions Centre (UAC). The findings and views reported in this essay are those of the author and should not be attributed to UAC.

This essay uses open source $\text{\LaTeX}$ and **Stata 17** packages including `reghdfe`, `tictoc`, and `ftools`.

DEDICATION

To my late grandfather Professor Ken Pearson; the original policy provider.

ABSTRACT

This paper investigates supply and demand effects that resulted from a legislated change in prices and subsidy levels across different areas of study at Australian universities. Using a conditional logit model with unit-record applications data, I find that university applicants are not highly price elastic. This is possibly due to generous interest-free study loans that substantially discount the present value of future debt repayments. On the supply-side, I analyse enrolment changes to estimate how responsive universities are to changes in marginal revenue by field-of-study. In the short run, I find that universities do not overtly reallocate supply of enrolments to courses that have higher per-student revenue after the policy change. This is possibly due to short-run supply constraints, reputation concerns, and other organisational priorities. I find that the policy increased the average annual cost of study by \$491, with higher increases for female and Indigenous students.

JEL classification: H52, I22, I23, I24, I28

Keywords: Student choice, university, supply, demand, elasticity

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1 Introduction

University enrolments are important in understanding the future of the labour market, particularly how productivity, expertise, and innovation are expected to grow. The disciplines in which today’s students choose to specialise impact the course of a country’s future. In Australia, policy-makers aim to ensure a healthy supply of healthcare practitioners, while concurrently nurturing an innovative science and technology cohort. So it is valuable to understand why university applicants pursue certain areas of study. In particular, how much do future students consider university fees in Australia’s system of *income contingent loans* (ICLs), whereby student debts are not repaid until a certain income threshold is met? Beyond student choice, it is also important to understand the role tertiary institutions play in determining the composition of each university cohort. If there is a shortage of teachers, nurses, or agricultural specialists, should policy makers attempt to persuade students or incentivise universities?

This paper addresses the question of how price sensitive students are in their study decisions, and how responsive universities are in capitalising on revenue-maximising allocations of courses. Specifically, this paper examines the effects of the 2021 ‘Job-ready Graduates’ policy that changed student fees and per-student university funding for enrolments in different fields of study. I use a conditional logit model with administrative applicant preference data to estimate how student demand varies in response to price changes. Then, I use a two-way fixed effects model with enrolment data to measure the university-side supply response.

The literature has not been able to consistently conclude whether students are deterred by higher prices in an ICL setting. The literature review in [Coelli \(2009\)](#) summarises how research in this area has variously estimated a negative pricing effect on demand, no effect, or an unclear mix of the two. In terms of the supply-side response, there is little literature analysing universities’ responses to changes in revenue per student.

The paper contributes to the literature by modelling individual student-level choice with rich unit-record preference data, which finds a weak demand response (controlling for student demographics). Moreover, this paper explores whether universities reallo-

cate supply in order to maximise revenue, finding a somewhat significant heterogeneous supply-side reallocation. To the best of my knowledge, it is the first to empirically analyse the Job-ready Graduates policy change. I also add to the literature of cost impacts for different demographic groups in their tertiary study decisions, finding that this policy disproportionately affected female and Indigenous students.

In this paper, [Section 2](#) provides an overview of the literature. [Section 3](#) contextualises the policy details that allow for a unique natural experiment. [Section 4](#) outlines the data and presents descriptive statistics. In [Section 5](#), I propose theoretical models for student choice and university supply allocation. Following the construction of appropriate econometric models in [Section 6](#), I present my results in [Section 7](#). [Section 8](#) discusses limitations and potential avenues for future research. [Section 9](#) concludes.

2 Literature Review

Tertiary education can be viewed as an investment in human capital, where private investment occurs when benefits outweigh costs. [Coelli \(2009\)](#) shows that a \$1,000CAD increase in Canadian university tuition reduced demand by 2.5 to 5 percentage points. However, the Australian context is markedly different due to ICLs, that have existed since 1989 ([Chapman, 1997](#)). There is some research on price elasticity in the context of Australian ICLs, with [Aungles et al. \(2002\)](#) finding that the introduction of HECS did not lower demand, but fee hikes in 1997 may have softened demand.¹ A consultant report that uses aggregate application data suggests that a series of price increases between 1997 and 2009 decreased demand ([Deloitte Access Economics, 2011](#)). However, the estimation approach is possibly confounded by various factors since the analysis only controls for school graduations and simple labour market statistics. Some non-economic policy research suggests that students are price insensitive ([Vossensteyn and Canton, 2001](#); [Higgins, 2019](#)).

¹Australian ICLs are called HECS-HELP (Higher Education Contribution Scheme and Higher Education Loan Program). When they were introduced in 1989, they were simply known as HECS.

Chapman and Khemka (2022) present a theoretical framework to understand the net present value of a HECS-HELP loan, specifically in the context of the Job-ready Graduates price changes. Their findings suggest that responses to the changes in price are muted under the assumption that university students heavily discount future HECS-HELP loan repayments.²

In the Netherlands, de Gendre and Kabátek (2021) find a lag effect in students' elasticity to tertiary education, which is posited to be the result of a twofold effect. Firstly, information lags increase student rationality as the policy becomes better known over time. Secondly, because high school students need to change subjects as early as three years pre-university to stream into specific degrees, the first few cohorts after the policy had a more constrained choice set. Moreover, Soutar and Turner (2002) found that students are concerned about more than price when selecting university degrees, taking into account course suitability, academic reputation, and job prospects as major decision factors.

While the literature suggests the broad majority of students are relatively price-insensitive as a result of income contingent loans, this paper explores whether the lowest percentiles of socioeconomic status (SES) are more price sensitive. Andrews (1999) finds that the introduction of HECS did not greatly deter students from studying at university, but is not conclusive on the composition of low SES applications. The paper introduces the hypothesis that low SES students are more debt-averse than other SES groups, suggesting that increases in HECS-HELP prices could deter applications despite no change in immediate financial burden. However, there is mixed evidence to support this hypothesis. Callender and Mason (2017) find that low SES UK students are more debt averse, however, this effect has decreased in the last decade. Cardak and Ryan (2009) find that a major confounder between SES and university enrolment is the Australian Tertiary Admissions Rank (ATAR) entry score, and once removed, there is a much weaker positive relationship.

²For example, using a conservative 5% discount rate, Chapman and Khemka (2022) calculate that for the average female studying Communications, a headline price increase of 113% would only factor as 83% in present value terms. This is because the loan repayments – which are incurred over decades – are not an imminent financial liability.

I am unaware of any published literature that explores the empirical results of the Job-ready Graduates reform. While there is nothing in the academic sphere, Andrew Norton has published blog posts using less rich aggregate enrolment data.³ Moreover, there is a small amount of theoretical work that critiques the policy design, rather than measure ex-post outcomes (Daly and Lewis, 2020; Koshy et al., 2020; Grant-Smith et al., 2020).

The vast majority of university supply-side analyses are focused on American colleges, which have the ability to set prices (Epple et al., 2006; Gansemer-Topf et al., 2021). In US colleges, there is some evidence of profit-motivated strategic reallocations in the context of contractionary funding changes (Cameron, 1983; Rolfe, 2003). In Australia, Avkiran (2001) explores university efficiency levels and production functions, but does not estimate responsiveness to revenue changes. Lindsay (1982) argues that universities are different from other institutions due to their “lack of profit motive, goal diversity and uncertainty, diffuse decisionmaking, and poorly understood production technology”, suggesting revenue-maximisation may not be a major concern.

3 Background

3.1 Income Contingent Loans and University fees in Australia

Income contingent loans (ICLs) are a special type of loan that require no upfront payment from university students until their income hits a threshold – currently \$48,361 (35th income percentile) in Australia.⁴ Repayment increases on a sliding scale, with the maximum repayment rate requiring those earning above \$141,848 annually (91st percentile) to repay 10% of their yearly income towards their HECS-HELP loan.⁵ Because the nominal interest rate is fixed to the Consumer Price Index (CPI), HECS-HELP debts do not accumulate real interest.

³See [Andrew Norton’s website](#) for more information.

⁴HECS threshold data from the [Australian Study Assist Website](#) for the 2022-23 financial year. Income percentile data from the [Australian Tax Office Taxation Statistics](#) for the 2019-20 financial year.

⁵HECS threshold data from the [Australian Study Assist Website](#) for the 2022-23 financial year. Income percentile data from the [Australian Tax Office Taxation Statistics](#) for the 2019-20 financial year.

In Australia, there are fixed prices and subsidies for each field-of-study, so per-student university revenue is given by:

$$\text{Revenue (per student)} = \underbrace{\text{Government Contribution}}_{\text{CSP Subsidy}} + \underbrace{\text{Student contribution}}_{\text{HECS-HELP Loan}}$$

When a student receives a Commonwealth Supported Place (CSP), the federal government pays for part of their studies through a *government contribution*.⁶ The student is expected to pay the remaining *student contribution*, with the majority of students (87.1%) opting to take out a HELP-HELP loan (ICL) to finance their payments.⁷ Under the assumption that a government is economically rational when it determines the funding structure, it should be concerned about the two prices that impact incentives:

1. Student contribution impacts students' decision making
2. Revenue (per student) affects universities' supply decisions

Technically, the 'price' each student faces is an individual-specific amount based on their repayment schedule and whether they repay the full loan before death (which are both determined by their future income horizon).⁸ The student contribution amount can be thought of as a *sticker price*, which is quoted to all students, but may not actually be the true incurred cost. I will refer to the student contribution amount as 'price' henceforth.

When HECS-HELP (then known as HECS) was introduced in 1989, there was a single price of \$1,800 per year for all undergraduate courses ([Chapman, 1997](#)). Under the Howard government in 1997, fees were differentiated by courses to align with expected income post-graduation ([Waghorne and Croucher, 2022](#)). Degrees associated with high expected incomes (e.g. law and economics) had higher \$5,500 annual fees, whereas low expected income courses (e.g. humanities and music) had lower \$3,300 annual fees.⁹ This

⁶A CSP is a per-student subsidy on fees, called the 'Commonwealth contribution' and referred to as the 'government contribution' in this paper. CSPs are generally available to all Australian citizens.

⁷See HELP Statistics on [Parliament of Australia website](#). Updated on 2nd of May, 2017.

⁸The 'true' price is unobservable, even to students themselves, since they do not have perfect information about their future income. Future research could consider a way to estimate an expected 'true' price based on observable student characteristics, which would allow for the calculation of a 'true' price elasticity.

⁹Fee clusters and a comprehensive history of HECS funding in Australia can be found on [The Australian Parliament website](#), titled "The Higher Education Loan Program (HELP) and related loans: a chronology". Updated on 23rd of March, 2021.

fee structure remained the general philosophy in place until Job-ready Graduates.¹⁰

3.2 The Policy Change

In 2020, the Australian Education Minister Dan Tehan announced the Job-ready Graduates Package: a new funding model for government-subsidised undergraduate university places. The aim of the legislation was to divert university applicants into study disciplines that would supposedly support the needs of the future labour market.¹¹ “Areas of national priority” including nursing, agriculture, and teaching were incentivised with lower fees.¹² On the other hand, “low need” areas including arts, communications, and law were disincentivised with higher fees.

The policy was announced on the 19th of June 2020, with the changes effective for the February 2021 enrolling university cohort. Existing university students had their fees grandfathered with no changes for the remainder of their degree – unless the cost for their course decreased in which case they began paying the lower fee from 2021.¹³

A highly criticised element of the policy design was the misalignment of incentives between students and universities presented in [Table 1](#) ([Daly and Lewis, 2020](#); [Koshy et al., 2020](#); [Grant-Smith et al., 2020](#)). For example, the government attempted to disincentivise enrolments in Criminology degrees, increasing the price by 117% (from \$6,684 to \$14,500). However, the per-student revenue that universities received for Criminology increased by 22% (\$12,800 to \$15,600), which would theoretically encourage universities to enrol more students. As universities set their own minimum entry score (ATAR), they have the ability to increase enrolments by lowering the entry score (assuming they have excess demand for that course). On the other hand, the government tried to increase Mathematics enrolments through a price decrease of 59% (\$9,527 to \$3,950). However,

¹⁰There were many fee changes between 1997 and 2020, however they all kept to the same general structure that aligned fees with expected future income. For a comprehensive review, see the above footnote.

¹¹Dan Tehan claimed “We are sending a price signal to encourage people to study in areas of expected employment growth”. See the [Ministerial Media Release](#) posted on 19th of June, 2020.

¹²For more detailed information, see ‘Job-ready Graduates Package’ on the [Department of Education website](#).

¹³With the exception of Honours students who had to pay the higher non-grandfathered fee due to an oversight in the legislation.

Table 1: Summary of Job-ready Graduates price and funding changes

Policy Change	Price change (student)	Revenue change (university)	Fields of Study
Cheaper & more funding	-17% -41%	4% 34%	IT, Architecture, Public Health Language
Cheaper & less funding*	-59% to -41% -29% to -17% -29% to -16%	-15% to -4% -31% to -28% -15% to -1%	Nursing, Teaching, Maths, Agriculture Medical Science, Environments Science, Engineering, Pharmacy, Optometry, Allied Health, Health (Broad)
More expensive & more funding*	1% to 19% 30% 51% to 117%	6% to 21% 17% 9% to 22%	Psychology, Medicine, Dentistry, Veterinary, Social Work, Creative Arts Business, Law, Economics Arts (Broad), Criminology
More expensive & less funding	67% 117%	-8% -11% to -22%	Creative Arts (Broad) Politics, History, Communications, Human Movement

Note: *Misaligned policy because the student and university incentives are misaligned. Broad categories are courses where the course costs are not obvious from enrolment, so a weighted average of expected units of study was used. ‘Revenue changes’ are the changes in the sum of the student and government contributions.

they also lowered per-student revenue by 15% (\$20,348 to \$17,200), which would encourage revenue-maximising universities to lower their supply of Mathematics degrees. I hypothesise that these mixed demand and supply incentives could lead to mixed effects on enrolments, which are a joint function of demand and supply.

4 Data and Descriptive Statistics

4.1 Data

The primary source of data is an administrative dataset from The Universities Admissions Centre (UAC). UAC is the sole clearinghouse tasked with matching applicants with all 26 tertiary institutions in New South Wales (NSW) and the Australian Capital Territory (ACT).¹⁴

¹⁴While UAC is the only matching centre, students can apply direct to universities and bypass UAC entirely. There is no data on the percentage of applicants who apply through UAC, but it is reportedly the vast majority.

Table 2: Fictitious applicant data

Year	Age	Gender	Leaver	Indigenous	SES	Postcode	Language	School	ATAR
2016	18	F	Yes	No	High	2043	English	Private	90-95

Preference	Course ID	Field of study	Course Start	Uni group	Level	Enrol?
1	5492	010101	Feb-16	Go8	Bachelor	No
2	5978	020101	Feb-16	Go8	Bachelor	Yes
3	122	020300	Mar-16	Metro	Bachelor	No
4	1679	060100	Feb-16	Metro	Bachelor	No
5	4189	010101	Feb-16	Regional	Bachelor	No

Note: ‘Leaver’ is a dummy variable for a high school applicant, and ‘Language’ captures the applicant’s primary language spoken at home. ‘School’ specifies whether the applicant attended a private or public high school. ATAR is given as a range, which is finer at higher ATARs. Age is calculated from year of birth data. SES is calculated from self-reported parental educational attainment data.

This analysis only focuses on Australian domestic applicants eligible for a government-subsidised CSP who are applying to Bachelor level courses. There are two years of post-policy observations (2021 and 2022), and the data captures application preferences and subsequent enrolments as seen in [Table 2](#). Due to data restrictions I aggregate universities into their ‘groups’, comprising three Group of Eight (Go8) universities (which account for 31.2% of enrolments), five Metro universities (44.3%), eight Regional universities (22.9%), and twelve ‘Other’ universities (1.6%).¹⁵ Applicant summary statistics are presented in [Table 3](#).

Table 3: Summary statistics for applicants 2014-2022

Variable	Applications	Enrolments
Female (%)	56.5%	55.2%
Age	21.2 (5.6)	20.9 (5.3)
English as primary language spoken at home (%)	71.6%	71.0%
Indigenous (%)	1.8%	1.5%
School leaver (%)	50.7%	43.2%
Number of students	736,026	371,890

Note: Standard deviations are in brackets.

For each de-identified applicant, the dataset provides information on parental educational attainment, which I will use as the primary indicator of SES. Also observable is the applicant’s postcode which is used to determine the SES of the applicant’s local

¹⁵UAC does not identify individual institutions due to their data privacy guidelines. The Go8 is comprised of University of New South Wales, University of Sydney, and Australian National University. A full list of university classifications has been provided in the supplementary materials.

community, referred to as Local SES henceforth.¹⁶

A six-digit field-of-study code allows me to identify all characteristics of the preferred degrees.¹⁷ Prices and subsidies for each year can be uniquely matched to each six-digit field-of-study code.¹⁸ However, since there are 237 different six-digit field-of-study codes in the UAC dataset, an aggregation approach is required for simplicity of interpretation. I define 27 fields of education that are unique by broad field of study (e.g. Science or Arts) and funding change post-policy.¹⁹

UAC use an applicant matching process that closely resembles the serial dictatorship (SD) algorithm, where the serial order is given by the ATAR score of each student (Artemov et al., 2017). As described by Abdulkadiroglu and Sonmez (1998), the SD matching process outlines that the student with the highest ATAR gets allocated their first preference course. The student with the next highest ATAR receives their first preference, which continues until a course reaches its enrolment limit (determined by the universities). If a student's first preference course is full, they get allocated their next highest-preference course that is not full.²⁰

I measure demand through applicants' first preference for a given field of study. Because each student is able to select up to five preferences (previously nine up until 2018), the matching algorithm is not *strategy-proof* (Haeringer and Klijn, 2009).²¹

¹⁶This measure is supposed to identify the effects of living in an advantaged or disadvantaged community, regardless of one's household wealth. This is determined by cross-referencing the applicant's postcode with the Australian Bureau of Statistics' [Index of Relative Socio-economic Advantage and Disadvantage \(IRSAD\)](#).

¹⁷The unique identifier for specific fields of study determined by Department of Education's [Australian Standard Classification of Education \(ASCED\)](#).

¹⁸Funding rates published on the [Department of Education website](#).

¹⁹There is slight funding heterogeneity in a few of the categories, however it is unlikely to make any material impact on this analysis. For example, the Politics & History category also includes Archaeology, Indigenous Studies, Religious Studies (and more) which had a slight increase in per-student revenue instead of the 11% decrease Politics & History faced. However these fields account for <5% of preferences and enrolments, so I have decided that it is worthwhile aggregating.

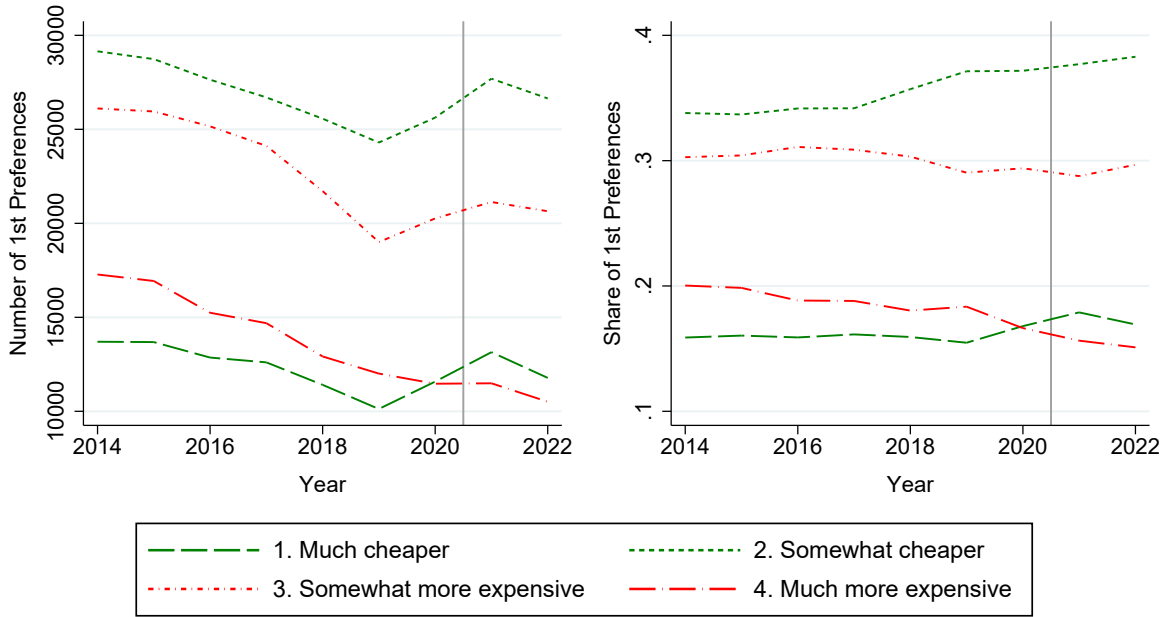
²⁰The UAC matching process is technically not a true SD algorithm. In actuality, universities set an ATAR entry threshold. So, students will receive an offer for their highest preference course where their ATAR is greater than the entry threshold. While setting this threshold, universities have access to full student preference data, so they can choose an ATAR threshold that perfectly matches their enrolment limit. It is useful to describe the process as an SD algorithm due to the non-strategy-proof property.

²¹A matching algorithm is *strategy-proof* if it is in the best interests of each applicant to report their true preferences (Nisan et al., 2007). Because UAC limits applicants to five preferences, the mechanism is not strategy proof, meaning that it may be in the best interest of some applicants to report non-truthful preferences.

Artemov et al. (2017) finds evidence in the Australian context that some applicants “adopt [preference] strategies that are unambiguously dominated”, meaning that applicants may not report truthful preferences.²² However, for the purpose of the logit demand model, incorporation of additional preferences would complicate identification and interpretation. While Fack et al. (2019) does suggest approaches to derive true preferences from dominated preference sets, it is out of the scope of this research. I analyse non-first preferences in Section 7.4.1 and reconsider the assumption that first preferences are a robust measure of demand in Section 8.

4.2 Descriptive Statistics

Figure 1: Demand (1st Preferences) by Price Change



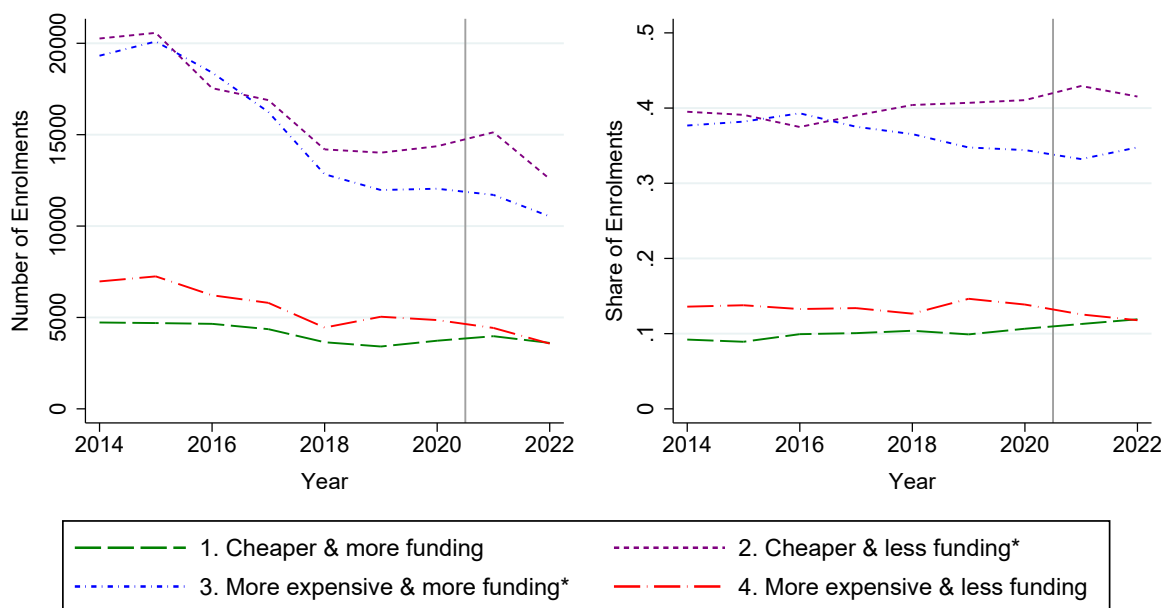
Note: ‘Much cheaper’ is when there is a price decrease of $>40\%$, and ‘Much more expensive’ is when there is a price increase of $>50\%$. I arbitrarily defined the categories to separate large and small price effects. The vertical line marks the policy change between 2020 and 2021.

In Figure 1, applicants’ first preferences are plotted over time to visualise demand. It is difficult to deduce a clear demand-side price response. Demand for degrees that became much more expensive ($>50\%$ price increase) did decline from 2020 to 2022. However, this could be a continuation of a trend from 2014 to 2020 which was led by lower demand

²²Artemov et al. (2017) suggests that this may be due to a lack of education on the UAC matching process (as well as the non-strategy-proof nature of the algorithm).

for Communications, Arts (Broad), Criminology, and Creative Arts (Broad).²³ Degrees that became somewhat more expensive ($<50\%$ price increase) faced higher demand post-policy, led by a recent uptick in preferences to study Psychology, Law, and Economics. While courses that became cheaper as a result of Job-ready Graduates experienced an increase in demand, it is difficult to determine whether there was a deviation from the pre-existing trend. Degrees that became much cheaper ($>40\%$ price decrease) saw a large increase in demand that began in 2019 prior to the policy introduction, with more preferences for Teaching and Nursing. There is evidence to suggest that the COVID-19 pandemic caused a worldwide increase in demand to study Nursing (Pfeifer, 2022; Bai et al., 2021). So, the increases in Nursing first preferences could be a result of a ‘taste’ shock, rather than a price change.

Figure 2: Enrolments by Price and Funding Change



Note: These categories correspond with those set out in Table 1. The large decline in group ‘3. More expensive & more funding*’ between 2015 to 2019 is mainly due to a fall in Business enrolments, which are currently 47% of their 2016 peak level. The vertical line marks the policy change between 2020 and 2021.

Figure 2 shows variation in enrolments over time, grouped into categories based on the direction of the change in student price and university revenue (same as in Table 1). Because enrolments are an equilibrium outcome co-determined by applicant preferences and university supply, it is difficult to interpret the mixed incentive funding categories

²³Broad fields exist when a student may take subjects from different funding clusters within a broad degree. I use the an enrolment-weighted average to calculate their expected cost.

(2 and 3). However, the ‘1. Cheaper & more funding’ category is straightforward to interpret since students and universities have aligned incentives to increase enrolments. There is a small increase in 2021 driven by Architecture, IT, and Public & Other Health, although it does dissipate in 2022. It is difficult to determine whether the increase in enrolments from 2020 to 2021 is a result of the policy since the upwards trend began in 2019 (before Job-ready Graduates was announced). Moreover, the COVID-19 pandemic likely had effects in 2021 and 2022, constraining universities’ ability to increase supply. The other category with aligned incentives is ‘4. More expensive & less funding’, which should theoretically decrease both demand and supply. While there is a decrease in enrolments post-policy (driven by Politics, History, and Human Movement), through visual inspection it is difficult to determine whether this is different from the recent trend. The inconclusiveness of the descriptive statistics suggest that a more rigorous econometric approach is required.

5 Economic Theory

5.1 Demand-side Theory

From the student demand perspective, this paper proposes a theoretical model for student course selection, conditional on already applying to university. I build upon the [Chapman and Khemka \(2022\)](#) net present value model by including non-financial factors that are important to student choice. Simply, the lifetime utility for individuals attending university is the sum of expected lifetime income and individual-specific benefits from studying their discipline of choice, less fees:

$$U_i(f_v) = Y_i(f_v) - C_i(f_v, Y_i(f_v)) + \gamma_i(f_v) \quad (1)$$

where U_i is lifetime utility, Y_i is the present value of expected lifetime income, γ_i is the weighted present value sum of all other factors (e.g. non-financial personal preferences) and C_i is the present value of the HECS-HELP loan. All factors are a function of f_v ,

the students' chosen field of study at university v . This model allows for heterogeneity in expected incomes from studying different disciplines at different universities. For example, studying medicine at a highly regarded university might be correlated with higher expected incomes than studying social work at a low-prestige university. C_i is also a function of income which is a key feature of income-contingent loans, since higher incomes require higher payments at time t , modifying the present value of costs. C_i is a function of f_v since courses have different student contribution amounts. For a representative 18-year-old university applicant, their lifetime utility function can be expressed as:

$$U_i(f_v) = \underbrace{\sum_{t=19}^{65} \frac{y_{it}(f_v)}{(1+r_i)^{t-19}}}_{\text{PV Income}} - \underbrace{\sum_{t=19}^{65} \frac{c_{it}(f_v, y_{it}(f_v))}{(1+r_i)^{t-19}}}_{\text{PV Costs}} + \underbrace{\gamma_i(f_v)}_{\text{Preferences}} \quad (2)$$

where students discount future income y_{it} and HECS-HELP loan repayments c_{it} at an individual-specific rate of r_i .²⁴ A rational applicant would aim to maximise their lifetime utility:

$$\max_{f_v} U_i(f_v) \quad \text{subject to} \quad ATAR_i \geq \underline{ATAR}_{f_v} \quad (3)$$

where in order to gain entry into a course, an applicant's ATAR needs to be greater than (or equal to) the minimum ATAR for that course at a particular university.

It is important to understand that a student's present value of course costs is necessarily lower than the quoted student contribution amount, or 'sticker price'. This is due to discounting over time, as well as the probability of not making repayments (which occurs if income is below the repayment threshold). Students' price sensitivity to their HECS-HELP loan is decreasing in their personal discount rate r_i , and in their time taken to repay. If r_i or their expected repayment time is sufficiently high, a change in fees will result in a small impact on lifetime utility. There is evidence to suggest that young adults have higher-than-average discount rates, so price sensitivity may be lower for younger applicants (Green et al., 1994; Read and Read, 2004).

Moreover, if the preference parameter $\gamma_i(f_v)$ is sufficiently high, it will outweigh any

²⁴The measure used for expected lifetime income y_{it} includes the probability of unemployment in that industry/discipline. The final working year is assumed to be at age 65 since that is when most Australians can access their retirement savings and/or state-funded pension.

cost concerns when choosing between courses. For example, an applicant who has always dreamed of becoming a lawyer would have a high $\gamma_i(f_v)$ that would dominate even a substantial increase in c_{it} . Although it would be difficult to estimate $\gamma_i(f_v)$, Soutar and Turner (2002) find that students have high preferences for quality teaching, campus atmosphere, and courses well-suited to them. Ultimately, this theoretical framework establishes the intuition for why students may not be price sensitive to university fees in the context of ICLs.

5.2 Supply-side Theory

Universities have a large number of strategic considerations when determining their optimal allocation of resources: maximising net revenue, increasing research impact, improving reputation, and more. At an undergraduate course level, it can be hypothesised that universities primarily focus on net teaching revenue. To capture this, I represent a university's net teaching revenue (NTR) function as following:

$$NTR_{cvt} = \underbrace{\sum_{c=1}^C G_{ct}E_{cvt} + \sum_{c=1}^C S_{ct}E_{cvt}}_{\text{Revenue}} - \underbrace{\sum_{c=1}^C (FC_{cvt} + MC_{cvt}E_{cvt} + MC_{cvt}^2E_{cvt}^2)}_{\text{Costs}} \quad (4)$$

where NTR_{cvt} is the sum of teaching revenue less costs for university v and course c , at time t . In revenue, G_{ct} is the government contribution and S_{ct} is the student contribution, which are taken as given across universities. E_{cvt} represents enrolments by course. In costs, FC_{cvt} are fixed costs and MC_{cvt} are marginal costs which follow a quadratic function.²⁵

When the Job-ready Graduates policy change occurred, there was significant variation in G_{ct} and S_{ct} , suggesting that it might be in universities' best interests to reallocate their supply of enrolments E_{cvt} . However, a key constraint on universities is the institution-specific Maximum Basic Grant Amount (MBGA) which places a hard cap on government

²⁵A quadratic function is used to represent a 'standard' marginal cost curve. Note that $FC_{cvt} = MC_{cvt} = 0$ for fields of study that are not offered in time t .

funding $\bar{G}_{vt}$.²⁶ Designed to stop universities from taking advantage of government funding by over-enrolling students, the MBGA acts as a disincentive to ensure universities limit their student load. For every additional enrolled student beyond the cap, universities receive only the student contribution S_{ct} (instead of $S_{ct} + G_{ct}$), which lowers marginal revenue by G_{ct} . Universities have been found to consistently breach this cap, which is financially optimal if marginal net-revenue is positive (marginal revenue from only the student contribution is still greater than marginal costs: $S_{ct} > MC_{cvt}$) (Norton, 2020). However, there are many other reasons why universities might over-enrol, including administration errors, pursuit of non-financial strategic goals, and to ensure that the government does not lower their MBGA in future years.

As a result of the policy, the maximum student contribution S_{ct} immediately grew 13% from \$11,155 to \$14,500, suggesting universities might increase over-enrolments in courses with a high S_{ct} . The MBGA places no restrictions on enrolment levels in each field of study, meaning universities are free to optimise supply between courses.²⁷ Because G_{ct} is a per-student amount that differs between fields of study and $\bar{G}_{vt}$ is a fixed dollar amount, additional enrolments in different courses will have varied marginal impacts on adding towards the total $\bar{G}_{vt}$ cap. So, universities who are at their MBGA limit will value marginal increases in student contributions S_{ct} more than increases in government contributions G_{ct} (since marginal revenue from G_{ct} is zero).

Following a funding change, it would be reasonable to expect a net-revenue-maximising institution to increase supply in (newly) high marginal revenue courses, and limit supply in lower marginal revenue courses. Formally, universities maximise net revenue:

$$\max_{E_{cvt}} NTR_{cvt} \quad \text{subject to} \quad \underbrace{\sum_{c=1}^C G_{ct} E_{cvt} \leq \bar{G}_{vt}}_{\text{MBGA limit}} \quad \text{and} \quad \underbrace{E_{cvt} \leq \min(\bar{E}_{cvt}^{Applicants}, \bar{E}_{cvt}^{Supply})}_{\text{Enrolment constraints}} \quad (5)$$

where the first constraint $\sum_{c=1}^C G_{ct} E_{cvt} \leq \bar{G}_{vt}$ represents the MBGA limit. The second

²⁶This is legislated under the [Higher Education Support Act \(2003\)](#). Each university's MBGA can be found on the [Department of Education website](#).

²⁷With the exception of Medicine degrees which have a field-specific funding cap as the result of being classified a 'designated higher education course'. See [Higher Education Support Act \(2003\)](#).

constraint $E_{cvt} \leq \min(\bar{E}_{cvt}^{Applicants}, \bar{E}_{cvt}^{Supply})$ states that the maximum number of students a university can enrol in each course is the minimum of the number of students who have applied to their institution $\bar{E}_{cvt}^{Applicants}$, and the number allowed within internal supply constraints $\bar{E}_{cvt}^{Supply}$ (e.g. Nursing hospital placements). Because universities set their own entry ATAR threshold, they can decrease the entry ATAR to increase enrolments.

Within this framework, it is clear that universities can re-optimize supply to favour courses with higher marginal net revenues. However, a drawback of this model is that it assumes universities are static one-period maximisers. This myopic analysis may abstract from important intertemporal considerations that universities might have, including the maintenance of reputation. Concerns around reputation might disincentivise universities from varying enrolment levels too quickly. Discontinuing a course, especially if it was for purely financial reasons, would likely impose a reputational penalty. On the other hand, there have been cases where Australian universities have suffered reputational damage for considerably increasing enrolments.²⁸ To increase enrolments, some universities decreased their ATAR entry scores significantly, which had reputational consequences partly because it signaled a low calibre of students.

6 Econometric Models

6.1 Demand-side: Conditional Logit

I specify a conditional multinomial logit model to identify applicants' responses to the price change. The model includes applicants' choices between 27 fields of study, where each of the 27 choices have unique price changes. Moreover, case variables allow the model to control for individual applicant attributes (e.g. gender and SES).

In an econometric context, the conditional logit model specifies that the utility of alternative j is a function of case-specific and alternative-specific regressors ([Hansen, 2022](#)):

$$U_j^* = W' \beta_j + X'_j \gamma + \epsilon_j \quad (6)$$

²⁸See [ABC Article](#) titled "Students with lowest ATAR scores being offered places in teaching degrees: secret report". Posted on 18th of September, 2018.

where W' represents the individual's case-specific attributes.²⁹ X'_j are the independent alternative-specific variables. I include student contribution and university revenue as X'_j in the model and omit the government contribution. University revenue is the sum of the student and government contributions, so I cannot include both revenue and the government contribution as it would result in perfect multicollinearity. To interpret the effect of a change in one variable, the other is implicitly held fixed. An increase in the student contribution holding the government contribution fixed would also increase university revenue. So (in this hypothetical model), the coefficient on student contribution might measure effects caused by universities' incentive to influence student demand. Specifying only student contribution and university revenue avoids this problem, and interpretation of each coefficient can be directly linked to incentives for students *or* universities. Note that the data does not observe any instances where one alternative-specific variable changes while the other one is held fixed. However, there is variation of the student contribution and university revenue in either direction, so both effects are identified.

Under the assumption that errors ϵ_j are distributed i.i.d. Type I extreme value, the probability response function is:

$$P_j(w, x) = \frac{\exp(w'\beta_j + x'_j\gamma)}{\sum_{l=1}^J \exp(w'\beta_l + x'_l\gamma)} \quad (7)$$

Letting $\theta = (\beta_1, \dots, \beta_J, \gamma)$, the log-likelihood function is:

$$\ell_n(\theta) = \sum_{i=1}^n \sum_{j=1}^J \mathbb{1}\{Y_i = j\} \log P_j(W_i, X_i|\theta) \quad (8)$$

The maximum likelihood estimator $\hat{\theta}$ maximizes $\ell_n(\theta)$. Since there is no closed form solution, $\hat{\theta}$ needs to be computed numerically. This can be performed using a [McFadden \(1973\)](#) conditional logit command on an econometric software ([Cameron and Trivedi, 2022](#)).

²⁹Case variables are comprised of gender, age, SES, ATAR, Indigenous status, school leaver status, language spoken at home, high school type, local SES, and year applied (to account for time trends).

6.2 Supply-side: Two-way Fixed Effects

To analyse the supply response, I identify the change in enrolments as a result of variation in revenue-per-student. Fixed and marginal costs are not observed, so I assume that changes in costs are not correlated with the policy change. I employ a two-way fixed effects (TWFE) model with the following specification:

$$\log(Y_{fgt}) = \beta_0 + \beta_1 \log(Rev_{fgt}) + \eta_{fg} + \gamma_{gt} + u_{fgt} \quad (9)$$

where the dependent variable Y_{fgt} is enrolments for a given field-of-study f and university group g , at year t . Rev_{fgt} is per-student revenue, which is the sum of the student and government contribution. To control for different levels of enrolments in different fields-of-study across university groups, I add fixed effects for fields-of-study and university groups η_{fg} . Then, to control for heterogeneous growth trends across the different university groups, I add university-group and time (year) fixed effects γ_{gt} .

While it would be ideal to control for different field-of-study enrolment trends with a time and field-of-study fixed effect, this would absorb all of the variation in the data, so I do not include it. A linear time trend for university-group and field-of-study $\eta_{fg} \times t$ is added in robustness checks in [Section 7.4.2](#). The error term u_{fgt} is assumed to be conditional mean zero, which rules out bias from unobserved confounders that affect enrolments and are correlated with revenue changes. In the most advanced version of this specification, I disaggregate revenue into the student contribution and government contribution, and I include university-group dummy variable interaction terms.

7 Results

7.1 Main findings: Demand

7.1.1 Full model

The results of the conditional logit estimation can be found in [Table 4](#) (heterogeneous results in Columns 2-8 will be explored in the next section). I am unable to use the

full dataset due to computational constraints, so I draw random samples of the data before estimating the model.³⁰ The alternative-specific variables X'_j are the logarithms of student contribution and university revenue. All specifications include case-specific variables W' , which are held constant when interpreting the effect of changes in X'_j .³¹

Table 4: Conditional Logit Demand Results

	(1)	(2)	(3)	(4)
	Full Model	Low SES	Medium SES	High SES
log(Student Contribution)	-0.024 (0.036)	-0.015 (0.054)	-0.033 (0.030)	-0.031 (0.049)
log(University Revenue)	0.008 (0.102)	0.054 (0.163)	-0.090 (0.089)	-0.021 (0.133)
% Random Sample	10%	50%	50%	10%
N cases	71,672	33,294	100,478	39,544

	(5)	(6)	(7)	(8)
	School Leaver	Non-leaver	English	Non-English
log(Student Contribution)	-0.028 (0.047)	-0.038 (0.041)	-0.002 (0.041)	-0.040 (0.033)
log(University Revenue)	0.195 (0.132)	-0.045 (0.118)	0.006 (0.120)	0.150 (0.085)
% Random Sample	10%	20%	10%	55%
N cases	35,911	71,454	51,218	113,231

[†] $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. *Note:* The full sample sizes were not used due to computational limitations. ‘English’ and ‘Non-English’ refers to whether the applicant’s primary language at home is English or not. ‘Leaver’ and ‘Non-leaver’ refers to whether the applicant is applying straight out of high school or not.

In the full model (1), the coefficient estimates for both student contribution and university revenue are statistically insignificant (p -values of 0.50 and 0.94 respectively). Note that conditional logit coefficients have no direct magnitude interpretation. The sign interpretation is that a negative coefficient suggests demand decreases as price increases. While the student contribution point estimate of -0.024 (an odds ratio of 0.976) does suggest some level of price sensitivity – whereby courses with a higher price post-policy face a fall in demand – the standard error causes it to be insignificant. Thus, it seems that students are essentially price insensitive. This concurs with [Christofides et al. \(2001\)](#)

³⁰ Attempting to use the full dataset in the full model resulted in the computer running for two weeks before crashing. In future research, I plan to use a supercomputer equipped with parallel processing capabilities.

³¹ The case-specific variables are gender, SES, Local SES, Indigenous status, school leaver status, English-speaking at home status, school type, ATAR, age, and year. This is except for the demographic specific models (2)-(8), where the heterogeneous effect being tested is not included in W' . For example, for the SES Regressions (2)-(4), SES is not included.

and the Neill (2009) probit model in the Canadian context.

The statistical insignificance of ‘university revenue’ is intuitive since revenue is unlikely to directly affect applicants’ decision-making. However, it is possible that over time this may have a significant positive coefficient if universities are able to mobilise marketing campaigns to increase demand in courses that provide them with higher revenue per student. Moreover, if applicants interpret higher university revenue as a signal for higher course quality, it could result in positive coefficient.

The framework set up in Section 5.1 shows why price is unlikely to be a major determinant for many students. Firstly, the net present value of university fees are much lower than the quoted nominal amount since graduates pay off their HECS-HELP debt over an average of 9.4 years.³² So, when an (average) applicant is determining their degree choice, they are discounting repayment over 12.4 years (including three years of study). Depending on their personal discount rate – which could be quite high for young people who are highly focused on their present finances – their present value of university fees can be much lower than the ‘sticker price’. In 2022, the price difference between the cheapest and most expensive course was \$10,645 per year. For an applicant choosing a 3-year degree that has a 7% discount rate and expects to earn the median Australian income after graduation, the present value difference becomes \$4,186 per year (61% less). This muted response to different HECS-HELP loan prices has been investigated by Chapman and Khemka (2022), who note that a price increase may have zero effect on applicants who expect to be in a low income percentile permanently, since they will never have to repay their debt.

Secondly, even if the present value of university fees is large enough to make some applicants price elastic, it is very small in comparison to expected lifetime income. Daly et al. (2015) estimates that Australian bachelors degrees have private rates of return between 8% to 58% (humanities and economics, respectively). So, applicants wishing to maximise lifetime income (assuming they have no preference parameter) will focus much more on expected income from courses, than the present value of the cost. Essentially,

³²Based on 2020-21 data in an [Australian Parliament HECS-HELP Loan Report](#).

the low present value of a HECS-HELP loan might be an almost-negligible consideration compared to expected lifetime income when an applicant is choosing their field-of-study.

Thirdly, it has been shown that students strongly consider non-financial factors when deciding what degree to study (Soutar and Turner, 2002). There is also literature that shows students take non-financial factors into account when determining the specific tertiary institution they want to attend (Johnston, 2010; Price et al., 2003). This implies that students may choose to study an otherwise non-preferred field of study in order to attend a university of choice, suggesting the existence of a non-negligible preference parameter $\gamma_i(f_v)$. The larger the value of this parameter, the lower the relative consideration of financial factors. So even a large nominal increase in university fees can result in a negligible response in student choice.

Overall, I am unable to find any price sensitivity in university applicants in response to the Job-ready Graduates price changes.

7.1.2 Demand-side Heterogeneity

I analyse heterogeneous demand-side responses by separately estimating subsets of data based on applicant demographics. This can be seen in Columns (2) to (8) in Table 4. As in the full model (1), there is no price sensitivity across different groups, suggesting no observable heterogeneity in the sample (p -values ranging from 0.22 to 0.96). Although none of the coefficients are significantly different from zero, the point estimates are all negative, possibly measuring a small amount of price sensitivity.

Andrews (1999) is unable to estimate whether university fee increases deter low SES students and posits that if low SES applicants are more debt-averse, they may be more price sensitive. My findings suggest that there is no difference for low SES applicants, which concurs with Christofides et al. (2001). Moreover, there is no observed heterogeneity between any SES groups, which might mean that price insensitivity is uniform across SES groups.

There is no significant heterogeneity between school leavers and non-school leavers. Non-school leavers are likely to face a higher present value of repayments relative to school

leavers since their income may already exceed the repayment threshold. If they need to make loan repayments sooner, they might be more price sensitive. However, this does not seem to be the case. While the point estimate suggests non-school leavers are more price sensitive, it is not significantly different from zero, which is the same for the school leaver coefficient. So, the higher present value of costs is likely to be of small consideration in comparison to income and personal preference considerations. Moreover, the non-school leaver cohort of applicants is likely to have some unobserved quality. The non-school leaver cohort includes those who are switching degrees as well as those who might have taken a ‘gap’ year or started working straight after high school.³³ Regardless, this does not seem to make a difference when measuring price sensitivity.

There is no observable difference between applicants from English and non-English speaking households. Strong culture pressures can be placed on students from non-English speaking households to study in areas with high prestige and high expected incomes, so it might make them less price sensitive (Somerville and Robinson, 2016). While the point estimate for suggests higher sensitivity for applicants from non-English speaking households, neither group’s sensitivity is significantly different from zero. Overall, there is no observable heterogeneity within the sample of NSW university applicants.

7.1.3 Re-specified demand model

In order to make use of the full data within computational constraints, I re-aggregate (unique) field-of-study groups into the four broad categories outlined in Table 1 and Figure 1.

While these groups can no longer be interpreted as discrete field of study choices, they are unique in direction of price and funding changes. The smaller model size allows for estimation of the full sample with the additional calculation of elasticities.³⁴ To compute this model, I need to aggregate the alternative-specific variables X'_j for each

³³Note that some students who take a ‘gap’ year (deferred tertiary study for one year) will apply in their final year of high school and defer the offer to enrol in the following year.

³⁴While a 10% sample of the 27-choice model was able to computationally converge, it took too long to calculate elasticities. After waiting for 3 weeks, I cancelled the process since I was unsure if or when it would finish.

Table 5: Funding changes in re-specified demand model

	<i>Percent change in</i>	
	Student Contribution	University Revenue
1. Cheaper and more funding	-17%	3%
2. Cheaper and less funding*	-26%	-13%
3. More expensive and more funding*	27%	14%
4. More expensive and less funding	116%	-7%

Note: *Misaligned policy because the student and university incentives are misaligned. Within each broad category shown, there are 4-11 unique fields of study. To calculate the funding change for each broad category, I take the average of each field-of-study funding change, weighted by 2020 enrolment levels.

‘choice’. Reported in [Table 5](#), I use the weighted average of the student contribution and university revenue amounts based on 2020 enrolment levels. Implications of this decision are discussed in [Section 7.4.1](#). It is important to note that each group only saw an increase *or* a decrease in student contribution and revenue, so although elasticities are reported, they have been calculated from data that only observed a price change in one direction.

Table 6: Elasticities using simplified course groupings

Model Specification	Student Contribution		University Revenue		n
	Elasticity	t-stat	Elasticity	t-stat	
Full Model	-0.0395***	-4.28	0.0359	1.40	628,502
Low SES	0.0216	0.66	-0.0507	-0.58	56,565
Med SES	-0.0192	-1.08	-0.0123	-0.25	173,321
High SES	-0.0576***	-4.76	0.0438	1.29	349,966
Leaver	-0.0473***	-3.96	0.1305***	3.82	336,038
Non-leaver	-0.0211	-1.41	-0.0846*	-2.11	292,464
English	-0.0296**	-2.82	-0.0405	-1.34	448,129
Non-English	-0.0726***	-3.72	0.2428***	4.88	180,373

[†] $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. *Note:* This was calculated by simplifying ‘course choice’ into four groups dependent on the student and university incentive changes outlined in [Table 1](#) and [Figure 1](#). The reported elasticity is the weighted average of the model’s four calculated elasticities, weighted by enrolments. There are missing observations for SES, so not all applicants fall into the three categories reported.

The results in [Table 6](#) show a significant non-zero elasticity in the full model ($p < 0.001$). The demand elasticity of -0.0395 suggests that a 1% increase in price is associated with a 0.0395% decrease in demand.³⁵ Conversely, a 1% decrease in price is associated with a 0.0395% increase in demand. So I can determine that students are not highly price sensitive, even if their elasticities are statistically non-zero. Measuring an elastic price

³⁵To reprise the discussion from [Section 3.1: Income Contingent Loans and University fees in Australia](#), in this model I measure elasticity with respect to the ‘sticker price’ referred to as the ‘student contribution’ amount. The ‘true’ price that each student faces is different based on their repayment probability and time horizon, which is a function on income. This cannot be observed, even by the student themselves.

response concurs with [Deloitte Access Economics \(2011\)](#) and [Aungles et al. \(2002\)](#), however, they do not quantify elasticity magnitudes so it is hard to directly compare results. The results suggest that school leavers, applicants from non-English speaking households and high SES applicants are the most price sensitive. However, the differences in elasticities are very small and the low and medium SES analyses have (comparatively) low power.

There are statistically significant results for some of the demand responses to university revenue. These are quite small and do not seem to be economically meaningful (this is discussed further in [Section 7.4.1](#)). They could capture successful marketing campaigns that encouraged school leavers to study disciplines that had high per-student revenue post-policy.

7.2 Main findings: Supply

7.2.1 Full model

Enrolments are the interaction between supply and demand. In [Section 7.1](#) I find that students are not highly price sensitive, so the likely consequence of a change in enrolments can be interpreted as a supply effect, assuming no time varying preferences or contemporaneous shocks.

The TWFE estimation in [Table 7](#) shows the sensitivity of universities to changes in revenue per student for different courses (heterogeneous university-group behaviour in Columns 2 and 4 is explored in the next section). In the first regression, the coefficient on revenue shows that there is no significant change in enrolments following an increase in per-student revenue (p -value of 0.16). In fact, the point estimate is negative, which could be interpreted as a decrease in enrolments for courses with higher funding levels (or an increase in enrolments for courses with lower funding levels). This is also possibly measuring a small demand-side response, since higher university revenue implies higher fees for students.

To disaggregate this, Regression 3 decomposes revenue into the student contribution and the government contribution amounts. Theory suggests that universities should more

Table 7: Equilibrium Supply effects

	Dependent Variable: Log Enrolments			
	(1)	(2)	(3)	(4)
log(Revenue)	-0.25 (0.17)			
log(Revenue) $\times$ Go8		-0.56 [†] (0.30)		
log(Revenue) $\times$ Metro		-0.45 (0.28)		
log(Revenue) $\times$ Regional		0.14 (0.30)		
log(Revenue) $\times$ Other		0.36 (0.57)		
log(Student Contribution)			-0.043 (0.097)	
log(Government Contribution)			0.034 (0.046)	
log(Student Contribution) $\times$ Go8				-0.046 (0.21)
log(Student Contribution) $\times$ Metro				-0.27 (0.22)
log(Student Contribution) $\times$ Regional				0.090 (0.17)
log(Student Contribution) $\times$ Other				0.51 (0.55)
log(Government Contribution) $\times$ Go8				0.059 (0.093)
log(Government Contribution) $\times$ Metro				-0.13 (0.093)
log(Government Contribution) $\times$ Regional				0.14 (0.098)
log(Government Contribution) $\times$ Other				0.36 [†] (0.18)
University Group $\times$ Field of Study FE	✓	✓	✓	✓
University Group $\times$ Year FE	✓	✓	✓	✓
Constant	5.04** (1.69)	4.79* (1.75)	2.69* (1.27)	2.73* (1.23)
Observations	10,579	10,579	10,579	10,579
Adjusted R^2	0.336	0.336	0.336	0.336

[†] $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. *Note:* Revenue is the sum of the student contribution and the government contribution. All dollar amounts are in real terms, indexed to 2022. Standard errors are clustered at the broad 27-group field of study level because that is the expected level of correlation.

highly value student contributions since they do not count towards the MBGA cap. The point estimates results show the opposite effect, although they are insignificantly different from zero and each other (F -statistic p -value of 0.208). The demand effect could be causing negative pressure on the student contribution coefficient, but it is not expected to affect the government contribution coefficient. It still may be true that universities value student contributions more, however the downwards pressure from a reduction in

demand could make the net effect insignificant.

A possible reason for why there is a weak observed supply-side response is that universities are relatively constrained to reallocate enrolments in the short run. Assuming that a university is at its MBGA cap, it might want to offset any increases in enrolments with lower enrolments in another course. Although Australian universities are notorious for having a highly casualised workforce, the lecturers and course coordinators who are integral for running degree programs are generally on a mix of medium-term contracts and ongoing (tenured) positions ([NTEU Policy and Research Unit, 2018](#)). Since it is difficult both financially and legally to expend this workforce in the short term, the marginal net-revenue may be higher for certain degrees even if the marginal revenue is lower. For example, if a university is obligated to pay its Mathematics faculty's wages for the next three years, even if revenue per student is lower than in the IT faculty, it may not be profitable to expend the Mathematics department and increase enrolments in IT. However, in the long run it would be unsurprising to see universities completely restructure their teaching focuses as a result of the Job-ready Graduates legislation. From a senior leadership perspective, universities may operate on a medium-to-long term horizon rather than focusing on year-to-year enrolments ([Lindsay, 1982](#)).

A major confounder in the data is the COVID-19 pandemic which greatly disrupted Australia between 2020 and 2022. In 2020 (when universities were deciding enrolments for 2021), there was a high level of uncertainty and a low level of confidence due to a negative shock in international student enrolments.³⁶ Given this high revenue risk, universities' senior leaders were possibly more focused on international student enrolments than relatively small gains available from taking advantage of domestic-student targeted legislation.

University reputation is likely an important factor for enrolment decision-makers. The financial intuition is that reputation increases the ability to generate revenue in the long run. There are examples of reputational damages that universities have faced as a result

³⁶See [The Conversation article](#) titled "COVID-19: what Australian universities can do to recover from the loss of international student fees". Posted on 3rd of June, 2020.

of cutting courses from their offerings.³⁷ So, a revenue-maximising university may not pursue gains by substantially reorganising its degree structure at the risk of losing more profit overall in reputational damage.

Lastly, universities may not have perfect information on their own marginal and fixed costs, making it difficult to maximise net teaching revenue.³⁸ Even if they do, the ‘optimal’ allocation for domestic undergraduate places may clash with the university-wide optimal allocation of resources.³⁹ Domestic undergraduate teaching revenue is unlikely to be the primary source of income for a university, so naïve optimisation of net teaching revenue NTR_{cvt} may not be the best university-wide strategy. Moreover, there may be information and agency issues that prevent university leadership from actually achieving the ‘optimal’ supply allocation.

7.2.2 Supply-side Heterogeneity

Regression 2 in [Table 7](#) shows a weakly significant heterogeneous supply-side response from different university groups. If I assume that demand elasticities are homogeneous across applicants for different universities, Go8 and Metro universities seem to exhibit behaviour converse to revenue-maximising since their relevant point estimates are negative (which is significant at the 10% level for the Go8 university group). This may be because of poor admissions planning, a reduced ability to reallocate supply, or a pre-existing growing focus on courses that happen to receive lower funding post-policy. Or, it could be measuring a negative demand effect as well. On the other hand, the point estimates for Regional and ‘Other’ universities seem to exhibit weak revenue-maximising behaviour, however, this is insignificant at a 5% level. The only (weakly) significant source of heterogeneity is between the Go8 and Regional university groups (F -statistic p -value of 0.090).

³⁷See [ABC article](#) titled “Major cuts at Charles Sturt University: Media courses merged, 116 degrees cut or changed”. Posted on 30th of July, 2020. Also, see [The Guardian article](#) titled “University of Tasmania to slash courses from 514 to 120 in wake of coronavirus impact”. Posted on 10th of March, 2020.

³⁸This could be because accounting practices might struggle to disaggregate domestic and international student costs, as well as research and teaching costs.

³⁹Realistically, reallocating enrolments would also reallocate research staff.

Regression 4 shows the decomposed revenue and heterogeneous university effects. Although all point estimates are insignificant at the 5% level, the positive response to an increase in government contribution for ‘Other’ universities is significant at the 10% level. Although insignificant, ‘Other’ universities seem to be more responsive to increases the student contribution compared to the government contribution, which is in line with supply-side theory. Go8, Metro, and Regional universities are the opposite, which again could be confounded by demand-side responses.

Table 8: Excess first preferences (demand) as a percentage of enrolments

	2020	2021	2022
Go8	131%	129%	182%
Metro	85%	115%	112%
Other	49%	49%	81%
Regional	139%	139%	177%
All universities	110%	123%	147%

Student demand is likely a key determinant of a university’s supply-side response. [Table 8](#) shows how Regional and Go8 universities enjoy the highest levels of excess demand, suggesting that they have a stronger ability to manipulate supply in order to increase revenue post-policy.

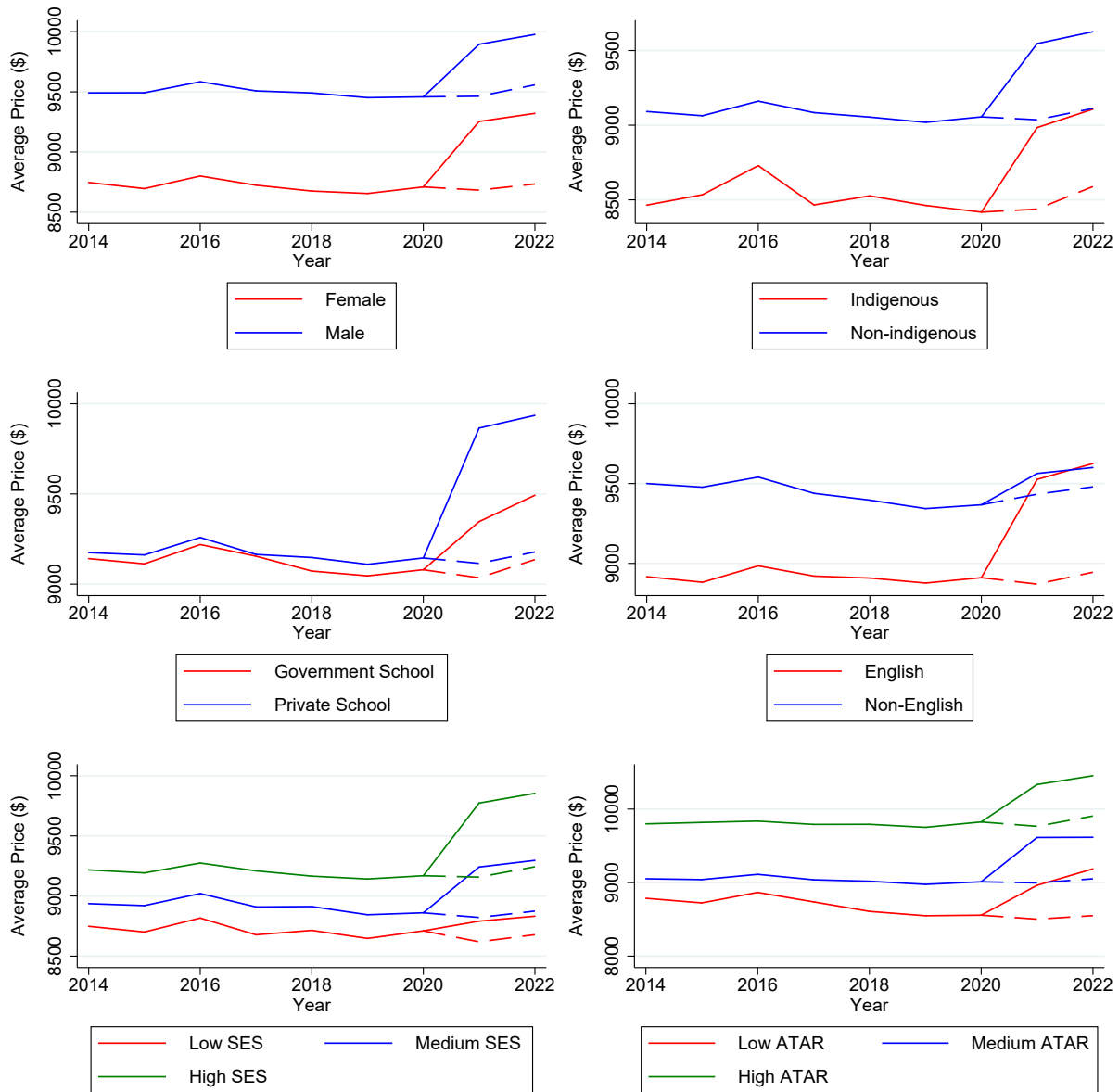
While larger universities often enjoy stronger reputations (and therefore more excess demand), they are likely subject to harsher reallocation constraints. Larger organisations may struggle to reallocate resources as quickly as smaller firms since they have more stakeholders and internal structural barriers.

7.3 Main findings: Cost Effects

To examine the policy’s effects on different demographics, I calculate the average annual cost for different groups before and after the policy, presented in [Figure 3](#). To account for changes in enrolments across fields-of-study, I calculate a counterfactual price change if the 2021 enrolment cohort (first cohort post-policy) received the old (CPI-adjusted) pre-policy prices. This allows me to determine the ‘price-only effect’, which is the direct result of the new prices without changes in enrolments across disciplines.⁴⁰

⁴⁰Calculated as the observed price minus the counterfactual price.

Figure 3: Cost Effects by Demographic



Note: Dashed lines are the counterfactual outcome based on CPI-adjusted pre-policy prices with post-policy enrolment levels. Dollar amounts are in real terms, indexed to 2022. The three ATAR groups are for ATARs less than 70, between 70-90, and above 90. School type, SES, and ATAR all have missing values in the dataset, so enrolments with missing data have been excluded from those relevant demographic calculations.

Women were worse off, since their observed increase in average yearly HECS-HELP costs post-policy was \$107.40 more than men. This is unsurprising since there is a higher proportion of female students in 11 of the 13 fields of study that faced price hikes. In 2021 and 2022, women made up between 60-67% of the Politics & History, Criminology, and Creative Arts cohorts, which all faced 117% price increases in the Job-ready Graduates legislation. While women paid less in fees than men before the policy, this is because they

tended to preference and subsequently enrol in courses with lower expected incomes.⁴¹ This did not change after Job-ready Graduates, suggesting that the policy lowered utility from studying for women (relative to men).

Post policy, Indigenous students' increase in average yearly HECS-HELP costs was \$75.72 more than non-Indigenous students. This higher relative cost burden on Indigenous students could worsen the wealth gap between Indigenous and non-Indigenous Australians. On average, Indigenous Australians' incomes are currently 66% the size of a non-Indigenous Australian's income (Markham and Biddle, 2018). Moreover, it is important to note that this analysis is conditional on university enrolment, which is already a large hurdle for many in the Indigenous community (Devlin, 2009).

Private school students faced an average \$719.78 increase in yearly fees post-policy, \$453.12 more than their government school counterparts. This is due to the fact that private school students generally make up the majority in the courses that increased which most in price (Human Movement, Politics & History, and Communications), and are in the minority for courses that decreased the most in price (Maths, Nursing, and Language). So, it seems the change in costs was due to a structural difference in courses that private students choose to enrol in compared to government school students.

Students who speak English at home faced a significant increase in costs compared to students from non-English speaking households. Following the policy, school leavers are paying \$122.1 more per year (on average).

In terms of SES, there is an equity-improving effect, where low SES students' increase in average fees is lower than medium and high SES students. The reason for the difference in average fees paid by different SES groups pre-policy is because high SES students tend to enrol in higher expected income courses.

The changes in different ATAR groups is mixed, with medium ATAR students' fees increasing the most post-policy.

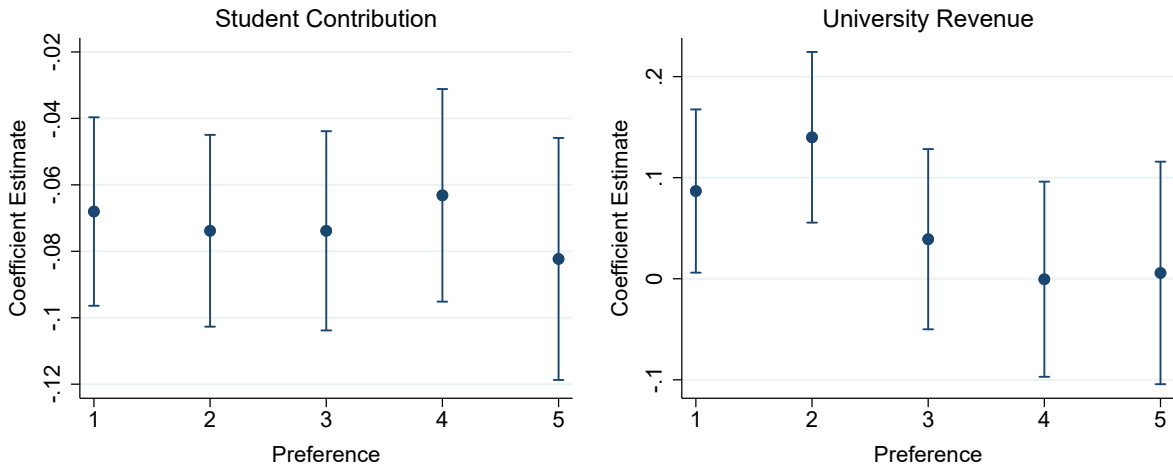
⁴¹This is because the old university fees were based on expected future income from studying in a certain discipline.

7.4 Robustness and threats to identification

7.4.1 Robustness of demand estimation

To analyse the robustness of the demand-side model, I analyse applicants' later (2nd to 5th) preferences. Using the simpler four-choice model from [Section 7.1.3](#), I estimate the impact of changes in demand in response to variation in the student contribution and university revenue. The 95% confidence intervals of the logit coefficient estimates are displayed in [Figure 4](#).

Figure 4: Logit estimates with different preferences



Note: 95% confidence intervals are reported. The sample sizes are smaller for each additional preference, since some applicants may not use all of their allocated preferences. While $n=628,502$ for 1st preferences, n falls until $n=335,453$ for 5th preferences.

While first preferences are relatively straightforward to interpret, subsequent preferences may represent a number of things. For example, a second preference may be a substitution away from a first preference. Moreover, students may have less interest in expressing their true intentions at lower preferences, since they may not expect them to eventuate. Despite this, there is a relatively constant small coefficient estimate for ‘student contribution’. Note that the first preference logit coefficient was used to calculate the elasticities reported in [Table 6](#). This suggests that the model is well specified, under the assumption that later preferences are still a positive indicator of demand, rather than a substitution effect from higher preferences.

The point estimates for ‘university revenue’ differ across preferences, with non-zero results for first and second preferences. The confidence intervals are all overlapping,

suggesting there may be no significant difference between different preferences. Future research could provide a more in-depth analysis of whether changes in university revenue affect student demand, possibly through marketing campaigns or signals of course ‘quality’.

A threat to identification is the specification of the four-choice model. In particular, the calculation of alternative-specific variables X'_j takes the averages of the student contribution and university revenue amounts, weighted by 2020 enrolment levels. As reported in [Table 5](#), the funding changes are a weighted average, meaning the model cannot determine differential effects from different magnitudes of funding changes. Moreover, this model does not capture student’s choices across fields of study, but only across different funding changes (i.e. higher or lower student contribution or university revenue post-policy). These limitations make the interpretation of the model more difficult. However, the statistically significant point estimate for price (-.068) in the four-choice model falls within the 95% confidence interval for price in the 27-choice model: [-.096 to 0.047]. This suggests that the four-choice model may acceptably identified.

7.4.2 Robustness of supply estimation

To test robustness of supply-side estimation, I add a university-group and field-of-study linear time trend, defined as ‘University Group $\times$ Field of Study $\times$ Year Trend’. This allows for each field of study at different university groups to have their own linear time trends. While a linear trend assumption is too strong for a baseline specification, it serves as an interesting comparison to the results provided in [Table 7](#).

The results of the new specification are reported in [Table 9](#). Compared to the baseline model, all coefficients in Regressions 1 and 2 have changed signs, but are still insignificant. In Regression 3, the coefficient on student contribution is now positive, but still smaller than the coefficient on government contribution.

In Regression 4, there is now evidence that Go8 universities strategically allocate enrolments to increase both student contributions and government contributions (assuming linear time trends). While these effects are statistically significant at the 5% level, their

Table 9: Equilibrium Supply effects with time trends

	Dependent Variable: Log Enrolments			
	(1)	(2)	(3)	(4)
log(Revenue)	0.061 (0.19)			
log(Revenue) $\times$ Go8		0.35 (0.29)		
log(Revenue) $\times$ Metro		0.19 (0.40)		
log(Revenue) $\times$ Regional		-0.25 (0.32)		
log(Revenue) $\times$ Other		-0.37 (0.83)		
log(Student Contribution)			0.080 (0.12)	
log(Government Contribution)			0.091 (0.060)	
log(Student Contribution) $\times$ Go8				0.33** (0.12)
log(Student Contribution) $\times$ Metro				0.015 (0.10)
log(Student Contribution) $\times$ Regional				0.030 (0.088)
log(Student Contribution) $\times$ Other				0.12 (0.21)
log(Government Contribution) $\times$ Go8				0.54* (0.22)
log(Government Contribution) $\times$ Metro				0.032 (0.23)
log(Government Contribution) $\times$ Regional				-0.15 (0.21)
log(Government Contribution) $\times$ Other				0.0089 (0.52)
University Group $\times$ Field of Study FE	✓	✓	✓	✓
University Group $\times$ Year FE	✓	✓	✓	✓
University Group $\times$ Field of Study $\times$ Year Trend	✓	✓	✓	✓
Constant	2.00 (1.91)	2.23 (1.82)	1.04 (1.62)	1.02 (1.38)
Observations	10,579	10,579	10,579	10,579
Adjusted R^2	0.340	0.339	0.340	0.339

[†] $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. *Note:* Revenue is the sum of the student contribution and the government contribution. All dollar amounts are in real terms, indexed to 2022. Standard errors are clustered at the broad 27-group field of study level because that is the expected level of correlation.

combined effect ‘log(Revenue) $\times$ Go8’ in Regression 2 is statistically insignificant (p -value of 0.24). This is likely due to the fact that the variation in the magnitudes of student and government contributions is higher than in their sum (university revenue). Although this coefficient is insignificant in this specification, in [Table 7](#), the point estimate is -0.56 with a p -value of 0.08. Further research could test the linear time trend assumption to determine which specification is most robust, which would be important in determining

the true heterogeneous effect.

Overall, a threat to identification is the low level of power in the supply-side analysis. Large standard errors make it difficult to determine whether the reported point estimates are measuring small effects or null effects. Moreover, the level of the variation in the data does not allow for a year and field-of-study fixed effect, which could theoretically bias results. However, without variation in price across universities, I am unable to control for this.

8 Limitations and Further Work

While all attempts have been made to correctly identify the effects of Job-ready Graduates, there are several potential limitations to the analysis presented in this paper. Firstly, a lack of computing power meant I was unable to run the full data sample for the rich 27-choice model in [Section 7.1](#). Future research could take advantage of parallel processing to estimate the model with all available data.

Secondly, the COVID-19 pandemic is a large confounder that cannot be controlled for since it perfectly coincided with the timing of the policy change. The data likely contains variation that is COVID-19 related and cannot be attributed to the policy change. For example, I calculate that the onset of the pandemic decreased the deferral rate by 73.5%, as many students planning to take a ‘gap’ year were pressured to study as lock-downs gave them few alternatives. The entirety of this paper’s analysis is impacted by the variation in student demand and university enrolments due to COVID-19.

Moreover, the timing of this research only allows for analysis of two post-policy periods. [de Gendre and Kabátek \(2021\)](#) find that when student demand changes in response to a pricing change, the effects are lagged, with larger effects each year post-policy until a new equilibrium is reached.⁴² So, there is a chance that Australian applicants will become measurably price sensitive in future years. Under similar logic, universities’ abilities to re-optimize supply of courses under internal budgetary pressures, contractual labour

⁴²This is partially due to the fact that university courses often have high school subject prerequisites, so it can be difficult to change one’s university plans in the final years of high school.

obligations, and long-term senior leadership strategies may be constrained in the short run. More measurable revenue maximising behaviour may become more apparent over the next few years as *factors of production* can be more easily reallocated. This lends itself to future research when more post-policy data is available.

The inability to directly observe individual universities limits the depth of this paper’s econometric analysis. Because universities optimise supply within-firm and are subject to a university-specific cap on funding, identification of firm behaviour would require individual institutions to be observed. The data in this paper is aggregated up to the university group to anonymise individual institutions which does not allow for a direct structural estimation approach. If individual universities are observed, further research could explore whether it is possible to analyse supply using a structural industrial organisation estimation approach based on [Berry et al. \(1995\)](#). A modified random coefficients logit model with fixed prices could potentially estimate each university’s marginal costs and their supply-side reallocation (if any) in response to Job-ready Graduates.

Another limitation is the assumption that applicants reveal true preferences when applying to universities through UAC. A behavioural study by [Guillen et al. \(2022\)](#) suggests that students may invoke a strategy that is different from their true preferences. However, this is an experimental study with a $< 5\%$ response rate which asks current university students to represent their preferences in a non-relevant counterfactual setting, so it may not accurately measure applicant behaviour. On the other hand, [Schnapp et al. \(2019\)](#) explores applicant preferences in a medical residency context, where the underlying matching system is the ‘stable marriage’ algorithm. They show that applicants may not always represent true preferences, however those who have been educated on the algorithm’s properties will. So it is feasible that UAC applicants who do not understand the matching process may not reveal their true preferences in a misguided attempt to improve their strategy. If many applicants play strategies that are not their true preferences, my research is caveated since I assumed that an applicant’s first preference is their sole measure of demand. However, if I interpret the first preference as a ‘revealed demand’, my analysis is robust.

Lastly, the use of UAC data means that empirical results only apply to the NSW population who have already decided to apply to a university. Firstly, the geographical constraints do not allow for heterogeneity in applicant behaviour between Australian states. While this is unlikely to be an issue when compared to a state of similar composition (e.g. Victoria or Queensland), the results may not directly translate to the Northern Territory or Tasmania. The Australian higher education system is essentially identical for all states (since it is legislated at a federal level), however there are significant differences in the high school systems across states. This could result in different student preferences. Also, UAC data constrains my analysis to those who have already chosen to apply to university. This means that I am unable to conduct any analysis into whether the policy affected students' decisions to attend university or pursue something else (e.g. TAFE, work, or a 'gap year').

While some of these limitations are difficult to circumvent (e.g. disentangling COVID-19 effects), others could be avoided by using an administrative dataset at a federal Australian level that identifies individual universities. While these exist at The Department of Education, they are rarely available to researchers.

9 Conclusions

From the analysis of the Job-ready Graduates university funding changes, I find that university applicants are not highly price sensitive in the context of income-contingent loans (ICLs). While estimation results vary between perfect inelasticity, and very low levels of price sensitivity, I find that price is not a key consideration for applicants. I contribute to the literature by analysing rich preference data, which is a much clearer indication of demand than enrolment data. This is in contrast to the majority of the existing literature that uses enrolments to derive demand, which can be confounded since enrolments are the equilibrium of supply and demand. The policy lesson is that financial incentives are not effective in modifying applicant behaviour. Moreover, there is evidence to suggest that non-financial factors (e.g. campus upgrades) could be more effective than

price changes.

Policy interventions to induce student demand could include the altering of wages in certain fields, or using high-school targeted marketing and education campaigns. For example, an effective way to increase applications to nursing degrees could be to transparently increase the hourly wage, especially in the first few years post-graduation.

On the supply-side, I find that universities do not act with overt revenue-maximising behaviour with respect to teaching revenue in the short run. With two periods of ex-post data, it seems that universities are not particularly responsive in capitalising on potential revenue gains. This is possibly due to internal constraints including reputation, long-term strategic visions, and imperfect information. There is weak evidence that Regional and ‘Other’ universities respond to increases in the government contribution. This is converse to theory, which suggests that student contributions are more valuable to universities.

If the government wants to control enrolments in specific fields-of-study that are vital for the future labour market, it does not seem pragmatic to enact supply-side or demand-side policy to elicit a short-term response. It is possible that universities will be more responsive to per-student revenue in the medium-term, as they recover from the COVID-19 shock and reallocate supply to improve their financial position. However, direct financial intervention seems to be ineffective in the short-term.

Overall, while evidence suggests that Job-ready Graduates is yet to meet its goals, it has allowed for an unprecedented analysis in student and university behaviour. The results of this paper may be insightful for policy makers, specifically in ICL education systems.

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